Jahnavi Menon

Quantum Software Engineer

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Work Experience

Bank of America **July 2024 – Aug 2026**

Software Development Engineer-1 *Hyderabad*

- Engineered and maintained a **secure credential-management system** syncing backend services and databases across applications, deployed and operated on **OpenShift**, and carried out **TLS certificate renewal** between applications — direct exposure to key-management and crypto-agility operations at enterprise scale.
- Designed and built a **Python + REST API automation service** for JIRA ticket creation across the team space, eliminating manual entry and saving **15+ hours/week** across 2 team members.
- Architected a **centralized database sync system (Python + PostgreSQL)** consolidating data from **multiple upstream sources in a high-throughput environment**, removing **5+ hours/week of manual reconciliation work**.
- Built a **Node.js release-automation service** adopted across 4 engineering teams, cutting release coordination overhead by **50%** and removing manual handoffs from the deployment pipeline.

Freelance and Open-source

Contractor **Aug 2026 – Present**

Software Development Engineer *Remote*

- Designing and developing an end-to-end application for standardized psychological assessments using **MERN and hybrid PQC encryption techniques** to ensure patient data privacy.

Quantum-Safe & Quantum Computing Projects

PriceWatch | *FastAPI, PostgreSQL, Redis, Docker, liboqs-python, Prometheus, Grafana, Locust, Pytest* **Source Code**

- Built a REST API with **PostgreSQL** and a **dockerized** multi-vendor scraping pipeline to track products, with **Prometheus** metrics and **Grafana** dashboards for service health; **Redis** caching, **Pytest** validation, and **Locust** load tests for throughput and cache-hit rates.
- Retrofitted the platform with a **post-quantum crypto-agility pilot**: a **hybrid X25519 + ML-KEM-768 key exchange** between services, deriving session keys via **HKDF** and authenticating price payloads with **AES-GCM**; **benchmarked hybrid vs. classical handshake** overhead under load.

QKD Encryption + Blockchain Storage System | *Python, Solidity, Web3, AES, Blockchain* **Thesis** | **Source Code**

- Designed a quantum-safe encryption layer by simulating **Quantum Key Distribution (QKD)**-inspired key-exchange logic in Python, layered on classical **AES** to explore hybrid classical/quantum-resilient design; encrypted records were stored in a **blockchain ledger** via **cryptographic hashing**.

Online Compiler Web App | *Docker, Express.js, MongoDB, REST API, OpenAI API* **Live Demo** | **Source Code**

- Built a **full-stack code execution platform** supporting **C++, Python, and Java** with an integrated AI chatbot (OpenAI API) for in-editor assistance.
- Implemented secure **JWT-based authentication**, user profiles, code persistence (MongoDB), and execution history tracking; containerized the app with **Docker** for isolated, reproducible execution.

Technical Skills

Quantum & Applied Cryptography: Post-quantum cryptography (ML-KEM/Kyber), hybrid key exchange (X25519 + ML-KEM), crypto-agility & crypto-inventory tooling, HKDF, AES-GCM, QKD simulation, public-key cryptography fundamentals, liboqs/oqs-python, Qiskit, QEC

Backend & APIs: Node.js, Express, Django, Flask, FastAPI, REST API design, Microservices

DevOps & Tools: Docker, Git, AWS, CI/CD, Postman, JIRA, Prometheus, Grafana

Languages: Python, TypeScript, Java, C++, Go, Bash

Databases & Caching: PostgreSQL, MongoDB, PL/SQL, Redis

Testing: Locust, Pytest, JWT-based auth testing

Awards and Volunteering

Silver Global Recognition Award **Dec 2025**

Bank of America

- Recognized for technical contributions across multiple lines of business in 2025.

Quantum Studies Focus Group (QSFG) Community **Sept 2026 – Present**

Co-Founder

- Started a study group, the Quantum Studies Focus Group (QSFG), with professionals from EY, startup founders, and students.

Education

Vellore Institute of Technology, Chennai **2020 – 2024**

B.Tech in Computer Science & Engineering *8.69/10 CGPA*

IBM – Basics of Quantum Information (Certificate of Completion) **Oct 2026**

Actively applying for the IBM Qiskit Advocate Program